CLASS 10 MATHEMATICS • CHAPTER 8

Introduction to Trigonometry

Exhaustive point-by-point presentation covering trigonometric ratios, specific angle values, identities, and applications directly from the textbook.

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Introduction & Real-World Right Triangles

🏛 Real-World Right Triangles

- **Qutub Minar:** Estimating height of tall structures without direct tape measurement.
- **River Width:** Determining river span from a temple balcony.
- **Hot Air Balloons:** Tracking altitude changes over time.

📖 Etymology & History

- Derived from Greek: **tri** (three), **gon** (sides), **metron** (measure).
- Study of relationships between sides and angles of a triangle.
- Origins in ancient Egypt, Babylon, and astronomical calculations.

Section 8.1 Live Check

Q1. What do the Greek roots tri, gon, and metron mean respectively in the word trigonometry?

Q2. Can you name a real-world monument or structure mentioned where right triangles can be imagined?

Q3. Why did ancient astronomers and early engineers rely heavily on trigonometry?

Defining the Six Trigonometric Ratios

Right Triangle Side Terminology (for acute angle A)

Hypotenuse (longest side opposite 90°), Side Opposite to angle A, and Side Adjacent to angle A.

sin A = (Opposite / Hypotenuse) | cos A = (Adjacent / Hypotenuse) | tan A = (Opposite / Adjacent)
cosec A = 1 / sin A | sec A = 1 / cos A | cot A = 1 / tan A = cos A / sin A

Section 8.2 Live Check

Q1. How is the sine of an acute angle A defined in terms of right triangle sides?

Q2. Which trigonometric ratios are reciprocals of sin A, cos A, and tan A respectively?

Q3. What is the relation expressing tan A in terms of sin A and cos A?

History of Trigonometric Terms & Symbols

🏛 Aryabhata & 'Sine'

- Aryabhatiyam (C.E. 500) introduced *ardha-jya* (half-chord), shortened to *jya* or *jiva*.
- Translated to Arabic as *jiva*, then to Latin as *sinus* (curve).
- Edmund Gunter (1581–1626) first used the abbreviation **'sin'**.

Cosine & Tangent

- Aryabhata called cosine *kotijya* (complementary sine).
- Edmund Gunter coined *cosinus*; Sir Jonas Moore (1674) introduced **'cos'**.
- Trigonometric ratio values depend only on the angle, not triangle size (by similar triangles).

Historical Note Live Check

Q1. What term did Aryabhata originally use for 'sine' in his work Aryabhatiyam?

Q2. How did the Latin word 'sinus' lead to our modern term 'sine'?

Q3. Do trigonometric ratio values change if you scale a right triangle up or down while keeping angles constant? Why?

Trigonometric Ratios of 30°, 45°, 60°, 0° & 90°

45° and 60°/30° Derivations

- **45°:** Isosceles right triangle with sides a, a, a√2. sin 45° = 1/√2, cos 45° = 1/√2, tan 45° = 1.
- **30° & 60°:** Equilateral triangle of side 2a split by altitude a√3.

0° and 90° Limits

- **0°:** sin 0° = 0, cos 0° = 1, tan 0° = 0, cosec 0° & cot 0° not defined.
- **90°:** sin 90° = 1, cos 90° = 0, tan 90° not defined, sec 90° not defined.

Specific Angles Live Check

Q1. What are the exact values of sin 30° and cos 60°? What do you notice about them?

Q2. Why is tan 90° not defined mathematically? Which side length approaches zero in the ratio?

Q3. As angle A increases from 0° to 90°, what happens to the value of sin A? Does cos A increase or decrease?

Fundamental Trigonometric Identities

What is an Identity?

An equation involving trigonometric ratios true for all valid angle values involved.

1. $\cos^2 A + \sin^2 A = 1$ (for $0° \le A \le 90°$)
2. $1 + \tan^2 A = \sec^2 A$ (for $0° \le A < 90°$)
3. $\cot^2 A + 1 = \operatorname{cosec}^2 A$ (for $0° < A \le 90°$)

Identities Live Check

Q1. What distinguishes a trigonometric identity from a conditional trigonometric equation?

Q2. State the Pythagorean trigonometric identity linking sec A and tan A.

Q3. Why is $\operatorname{cosec}^2 A = 1 + \cot^2 A$ not defined for $A = 0°$?

Worked Exercise Highlights

Problem 1: Given tan A = 4/3 (Ex 8.1 Q1)

- Let BC = 4k, AB = 3k. Hypotenuse AC = 5k.
- sin A = BC/AC = 4/5.
- cos A = AB/AC = 3/5.
- **All other ratios derived accordingly.**

Problem 2: Identity Proof (Ex 8.4 Q4(ii))

- Prove: $\cos A / (1 + \sin A) + (1 + \sin A) / \cos A = 2 \sec A$.
- Take LCM: $\cos^2 A + (1 + \sin A)^2 / [\cos A(1 + \sin A)]$.
- Expand numerator: $\cos^2 A + 1 + 2 \sin A + \sin^2 A = 2 + 2 \sin A = 2(1 + \sin A)$.
- **Cancelling gives $2 / \cos A = 2 \sec A$. Proved!**

Exercise Highlights Live Check

Q1. When given a single trigonometric ratio like tan A = 4/3, why do we introduce a positive multiplier k when finding side lengths?

Q2. What algebraic strategy (like taking LCM or rationalising) is most commonly used when proving fractional trigonometric identities?

Q3. Can sin theta ever exceed 1 in a right-angled triangle? Why or why not?

Chapter Recap & Checklist

Basic Definitions

- **Trig Ratios:** sin, cos, tan, cosec, sec, cot defined via right triangle side ratios.
- **Specific Angles:** Exact values for 0°, 30°, 45°, 60°, 90°.

Identities & Bounds

- **Identities:** $\sin^2 A + \cos^2 A = 1$, $1 + \tan^2 A = \sec^2 A$, $\cot^2 A + 1 = \operatorname{cosec}^2 A$.
- **Bounds:** sin A & cos A never exceed 1; sec A & cosec A ≥ 1.

THANK YOU

"There is perhaps nothing which so occupies the middle position of mathematics as trigonometry." — J.F. Herbart

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